

Mattia Martini

Postdoctoral researcher

Contact info

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Experience

Postdoctoral researcher

Sept 2025 – now

Centre de Mathématiques Appliquées (CMAP), École polytechnique

- Mentors: Mao Fabrice Djete and Huyên Pham
- Part of the ANR PEPR project MIRTE

Postdoctoral researcher

Apr 2023 - Aug 2025

Laboratoire J.A. Dieudonné (LJAD), Université Côte d'Azur

- Mentor: François Delarue
- Member of the ERC project ELISA

Education

PhD Mathematical sciences, Università degli studi di Milano

Oct 2019 – Mar 2023

- Thesis: *Advances in stochastic analysis on spaces of measures: Kolmogorov equations related to stochastic filtering and mean field optimal stopping*
- Supervisor: Marco Fuhrman

MSc Mathematical engineering, Politecnico di Milano (110/110 cum laude)

Sept 2017 – Oct 2019

- Thesis: *Kolmogorov Equations on Space of Measures Associated to Stochastic Filtering Processes*
- Supervisors: Sandro Salsa and Marco Fuhrman

BSc Mathematical engineering, Politecnico di Milano (110/110)

Sept 2014 – Sept 2017

- Thesis: *Quantum bits and superdense coding*
- Supervisor: Matteo Gregoratti

High school diploma, I.T.I.S Galileo Galilei, (100/100 cum laude)

2009 – 2014

Publications

Preprints

- [1] M. F. Djete and M. M. “W-BSDEs with Common Noise: Well-posedness, Characterization, and Consequences”. arXiv:2610.06593. 2026.
- [2] M. F. Djete and M. M. “A McKean–Vlasov semigroup and its application to a martingale representation problem”. arXiv:2608.03412. 2026.
- [3] F. Delarue, M. M., and G. E. Soudini. “HJB equations driven by the Dirichlet-Ferguson Laplacian in Wasserstein-Sobolev spaces”. arXiv:2511.03522. 2025.
- [4] A. Cosso, M. M., and L. Perelli. “Mean convergence rates for Gaussian-smoothed Wasserstein distances and classical Wasserstein distances”. arXiv:2504.17477. 2025.

Published papers

- [5] A. Cecchin, S. Daudin, J. Jackson, and M. M. “Quantitative convergence for mean field control with common noise and degenerate idiosyncratic noise”. *Electron. J. Probab.* 30 (2025), pp. 1–62.
- [6] G. Franzese, M. M., G. Corallo, P. Papotti, and P. Michiardi. “Latent Abstractions in Generative Diffusion Models”. *Entropy* 27.4 (2025).
- [7] F. Delarue and M. M. “Fourier Galerkin approximation of mean field control problems”. *Nonlinear Differential Equations and Applications NoDEA* 32.1 (2024), p. 11.
- [8] M. M. “Kolmogorov equations on the space of probability measures associated to the nonlinear filtering equation: the viscosity approach”. *Stochastic Analysis and Applications* 42.6 (2024), pp. 987–999.
- [9] A. Cosso and M. M. “On smooth approximations in the Wasserstein space”. *Electronic Communications in Probability* 28 (2023), pp. 1–11.
- [10] B. Djehiche and M. M. “Time-inconsistent mean-field optimal stopping: A limit approach”. *Journal of Mathematical Analysis and Applications* 528.1 (2023).
- [11] M. M. “Kolmogorov equations on spaces of measures associated to nonlinear filtering processes”. *Stochastic Processes and their Applications* 161 (2023), pp. 385–423.
- [12] M. M. and G. E. Soudini. “Numerical Methods for a System of Coupled Cahn-Hilliard Equations”. *Commun. Appl. Ind. Math.* 12 (2021), pp. 1–12.

Teaching

Stochastic Calculus and Mathematical Finance , Università degli Studi di Torino	2026
Lecturer (25h) MSc Quantitative Finance and Insurance	
Stochastic differential equations , Politecnico di Milano	2022
Teaching assistant (30h) MSc Mathematical Engineering	
Stochastic differential equations , Politecnico di Milano	2021
Teaching assistant (30h) MSc Mathematical Engineering	
Basics of Statistics , Politecnico di Milano	2021
Teaching assistant (30h) BSc Biomedical Engineering	
Stochastic differential equations , Politecnico di Milano	2020
Teaching assistant (30h) MSc Mathematical Engineering	
Probability and Statistics , Politecnico di Milano	2020
Teaching assistant (22h) BSc Management Engineering	

Given talks, seminars and posters

Demi-journée MIRTE , Institut Louis Bachelier Seminar of the ANR PEPR MIRTE	Sept 2026
Fifth Italian Meeting on Probability and Mathematical Statistics , Università di Palermo In the session "Stochastic Systems, Mean-Field Models, and Control"	June 2026
Torino seminar series in Stochastics and Mathematical Statistics , Università di Torino Seminar of the probability and statistics group	Apr 2026
WS Mean-field systems in finance, neurosciences and AI , Folgarida, Trento Contributed talk	Jan 2026
Seminario di Probabilità e Statistica , Politecnico di Milano Seminar of the probability and statistics group	Jan 2026
Seminari in Probabilità e Finanza , Università di Padova Seminar of the probability and finance group	Dec 2025
Séminaire du Pôle Probabilités , École polytechnique (CMAP) Seminar of the probability group	Oct 2025
Interacting Particle Systems and Applications , Università di Siena Invited talk	Sept 2025
16th Viennese Conference Optimal Control and Dynamic Games , TU Wien In the invited session "Stochastic Control and Applications"	July 2025
Durham Symposium on Mean Field Games , Durham University Invited talk	March 2025
WS Optimal Transport: from robust pricing to model calibration , Folgarida, Trento Contributed talk	Feb 2025
Probability and Statistics Seminar , University of Luxembourg Seminar of the probability and statistics group	Jan 2025
PDE Seminar , Georgia Tech (online) Seminar of the PDE group	Sept 2024
2nd AMS-UMI International Joint Meeting , Università degli Studi di Palermo In the invited session "Mean Field Games and related topics"	July 2024
Seminario di Probabilità e Statistica , Politecnico di Milano Seminar of the probability and statistics group	June 2024
Fourth Italian Meeting on Probability and Mathematical Statistics , Roma In the invited session "Stochastic optimal control of McKean-Vlasov equations"	June 2024
Young Researchers in Stochastic Control and Games , Universität Bielefeld Invited talk	Dec 2023
Séminaire de Probabilités et Statistique , Université Côte d'Azur (LJAD) Seminar of the probability and statistics group	Dec 2023
PGMO Days , EDF Lab Paris-Saclay In the invited session "Mean field games and control"	Nov 2023
A Random Walk in the Land of Stochastic Analysis and Numerical Probability , CIRM Luminy Poster	Sept 2023

SPDEs, optimal control and mean field games , Universität Bielefeld Poster	July 2023
The 9th International Colloquium on BSDEs and Mean Field Systems , IAE Savoie Mont Blanc Contributed talk	July 2022
Third Italian Meeting on Probability and Mathematical Statistics , Università di Bologna In the session "New trends in stochastic control"	June 2022
100 years UMI and First UMI meeting of PhD students , Università di Padova Contributed talk	May 2022
Seminars cycle "Insalate di Matematica" , Università degli Studi Milano-Bicocca Seminar organized by the PhD students	Nov 2021
Trento-Wuppertal School on Stochastic Processes, Analysis and Semigroups , Università di Trento Contributed talk	Sept 2021

Visiting periods

Università degli Studi di Torino , 1 Month Visiting professor position, ESOMAS department	Apr-May 2026
University of Luxembourg , 1 Week Collaboration with Michele Stecconi	Jan 2025
Imperial College, London , 1 Week Collaboration with Ofelia Bonesini	Jul 2024
Università di Parma, Italy , 1 Week Collaboration with Alessandro Calvia	Feb 2024
KTH, Stockholm , 1 Week Collaboration with Boualem Djehiche	May 2022

Organization

Contributed session "(Stochastic) Analysis on Spaces of Probability Measures and Applications" , with G. E. Sodini 5th Italian Meeting on Probability and Mathematical Statistics, 2026 Palermo
Workshop on Stochastic Processes, Stochastic Optimal Control, and their Applications , among the organizers September 26 - 27th, 2024 Politecnico di Milano
Contributed session "Mean Field Games and Mean Field Control I" , with J. Dianetti 4th Italian Meeting on Probability and Mathematical Statistics, 2024 Roma
Phd Days 2021 , among the organizers December 14 - 17th, 2021 Università degli Studi di Milano

Others

Abilitations: Maître de conférences (26 - Mathématiques appliquées et applications des mathématiques)

Referee activities:

- Annals of Applied Probability, Applied Mathematics and Optimization, SICON, Stochastic Processes and Their Applications, Journal of Differential Equations, Systems & Control Letters, Statistics & Probability Letters
- ANR (French grant)

Participation to research projects:

- Sept 2025 - now: part of the ANR PEPR project MIRTE (ANR-23-EXMA-0011)

- Apr 2023 - Aug 2025: member of the ERC ADV project ELISA (Grant agreement No. 101054746), P.I. F. Delarue.

Memberships:

- From 2025: Gruppo UMI Prisma (PProbability In Statistics, Mathematics and Applications)
- From 2022: Unione Matematica Italiana (UMI)
- From 2020: Indam, GNAMPA

Languages: Italian (Native) | English (Full professional proficiency) | French (Intermediate)

October 6, 2026
Mattia Martini